HANDWRITTEN STUDY NOTES

GEOSYNTHETICS AND REINFORCED EARTH STRUCTURES

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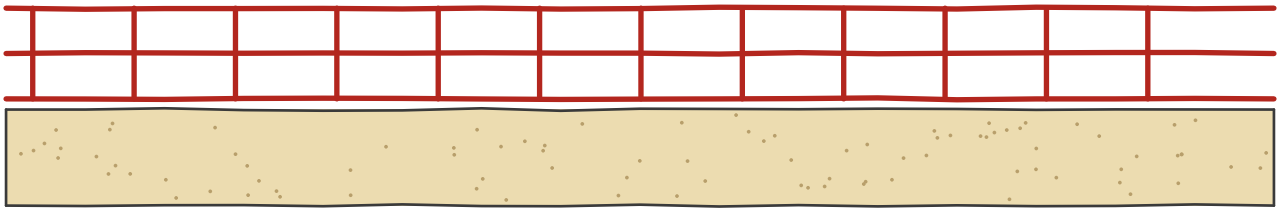
UNIT – V

Reinforced Soil Beds and Reinforced Pavements

Syllabus topics covered in these notes:

- Reinforced soil beds: introduction, factors affecting behaviour, analysis and design
- Reinforced pavements: benefits of reinforcement in flexible pavement layers
- Design of reinforced pavements by Giroud and Noiray approach
- Design of reinforced pavements by modified CBR method

soil + reinforcement = strong composite



1. Reinforced Soil Beds – Introduction

A **reinforced soil bed** (reinforced earth slab / reinforced foundation) is a layer of soil in which horizontal reinforcement (metal strips, geogrid, geotextile) is placed at several levels **below a shallow foundation** to increase its **bearing capacity** and reduce **settlement**. Pioneering work: **Binquet and Lee (1975)** with aluminium strips; later by Fragaszy and Lawton, Guido, Das, Khing, Omar and others using geogrids.

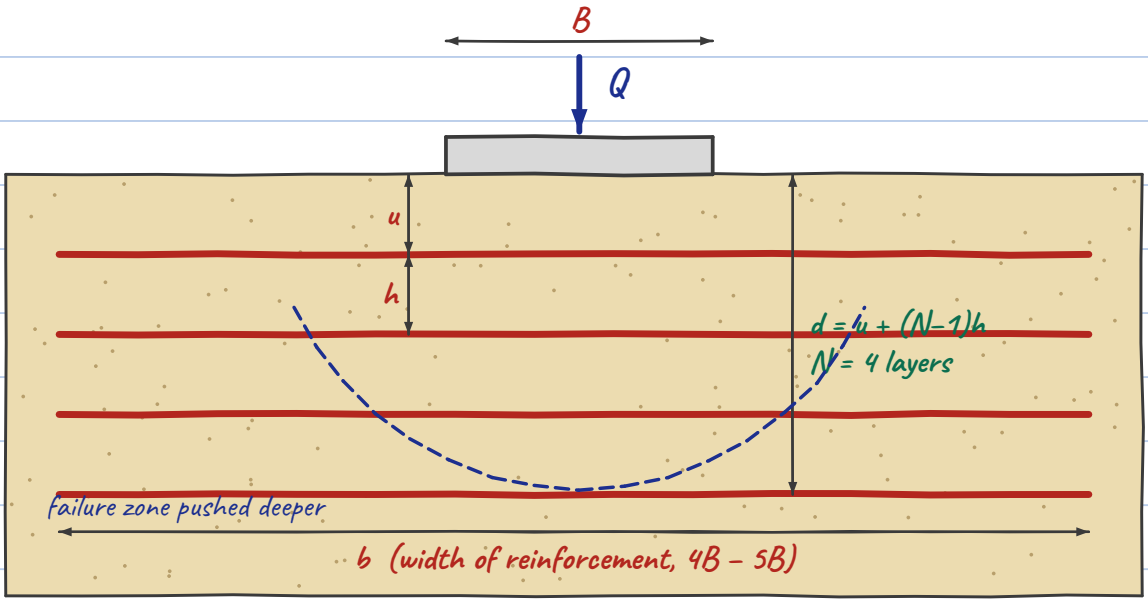


Fig 5.1 Footing on a reinforced soil bed

Geometric parameters

Symbol	Meaning	Typical optimum
B	width of footing	–
u / B	depth of first layer below footing	0.3 – 0.5 (≤ 2/3)
h / B	vertical spacing of layers	0.25 – 0.5
N	number of layers	3 – 5 (beyond: little gain)
b / B	width (length) of layers	4 – 5 (strip), 2.5–3 (square)
d / B	depth of reinforced zone d = u + (N–1)h	≈ 1.5 – 2

Mechanisms of improvement

1. **Deep-footing / rigid-boundary effect** – reinforced zone acts like a stiff slab, pushing the failure surface deeper into stronger or less stressed soil.
2. **Confinement / lateral restraint** – friction on the layers restrains lateral soil movement under the footing (enhanced confining pressure, pseudo-cohesion).
3. **Tensioned-membrane effect** – at large settlement the layers sag and their vertical component of tension carries part of the load.
4. **Load spreading** – wider and more uniform pressure on the lower soil.

2. Factors Affecting Behaviour of Reinforced Soil Beds

- **Depth of first layer (u/B)**: BCR is highest for $u/B \approx 0.3–0.5$; if $u/B > \approx 1$, the top layer is outside the stressed zone and gives almost no improvement (punching in the soil above).
- **Number of layers (N) and spacing (h/B)**: BCR increases with N, then levels off ($N \approx 4–5$); closer spacing is more effective but cost increases. Effective depth $d/B \approx 1.5–2$.
- **Length / width of reinforcement (b/B)**: must extend beyond the failure zone for pull-out resistance; BCR increases with b/B up to $\approx 4–5$ and then stays constant.
- **Type, strength and stiffness of reinforcement**: stiffer / stronger (steel strips, high modulus geogrid) gives higher BCR; too weak a layer **breaks**; geogrid aperture should match soil size (interlock).
- **Soil properties**: relative density and φ of the bed – benefit is higher for **loose sand** at small settlements; for very dense sand BCR is lower; cohesive soils need drainage layers.
- **Footing shape and size**: strip > square > circular in absolute gain; scale effect (small model footings overestimate BCR).
- **Surface roughness / interface friction** of reinforcement – higher friction, higher BCR.

- **Settlement level:** BCR is larger at larger settlement (s/B = 10–15 %) than at low settlement.

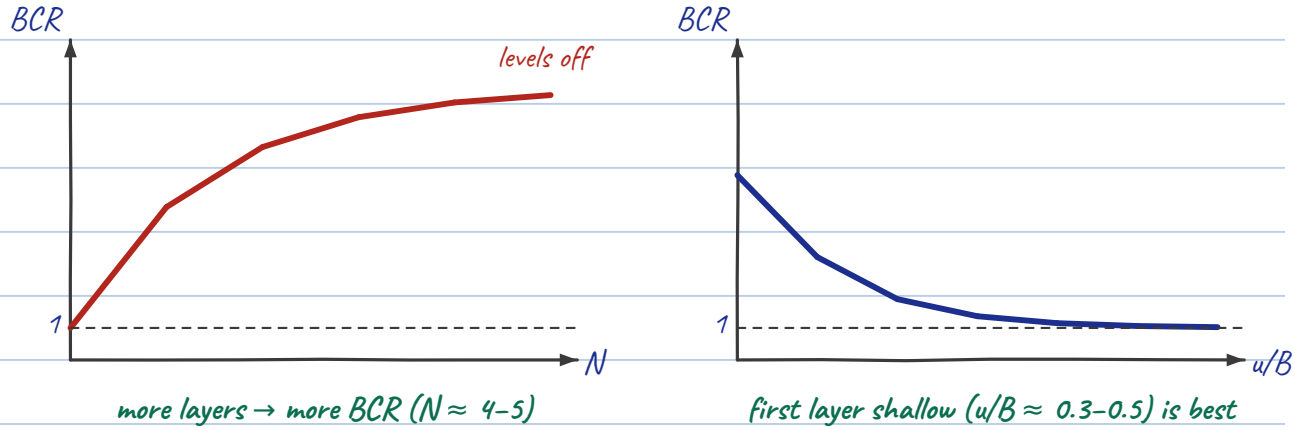


Fig 5.2 Typical trends of BCR (schematic)

Definitions

Bearing Capacity Ratio: $BCR = q_R / q_0$ (q_R = bearing pressure of reinforced bed, q_0 = unreinforced) at the same settlement or at failure.

Settlement Reduction Ratio $SRR = (\rho_0 - \rho_R)/\rho_0$ at the same load.

3. Analysis and Design of Reinforced Soil Beds

3.1 Failure modes (Binquet and Lee)

1. **Reinforcement breaks** (tension failure) – when layers are weak, many, and close to the footing (u/B small).
2. **Reinforcement pulls out** – when length is short or friction is low.
3. **Soil failure above the top layer** – when u/B is large (> 2/3): footing punches through the cover soil.
4. **General shear below the reinforced zone** – when d/B is too small.

3.2 Design procedure

1. Find q_0 (unreinforced) of the foundation soil: $q_0 = cN_c + qN_q + \frac{1}{2}\gamma B N_\gamma$, and the settlement it would give.
2. Choose the geometry: u/B = 0.3–0.5, h/B = 0.3–0.5, N = 3–5, b/B ≥ 4 (strip) → reinforced depth d.

3. Estimate **BCR** from design charts (Binquet and Lee) / test data: $q_R = BCR \cdot q_0$; $q_{all} = q_R / FS\ (\geq 3)$.
4. **Tension in layers:** compute from the load spread (T_{max} at the layer below the footing edge); check $T_{al} \geq T_{max}$ after reduction factors, with $FS \geq 1.5–2$ against breakage.
5. **Pull-out:** $L_e \geq FS \cdot T / (2 \mu^* \sigma_v)$ beyond the failure zone ($L_e \geq 1\ B$ typical); total width b = footing zone + $2 L_e$.
6. Check **settlement** (SRR) and the strength of the underlying soil at depth d (stress spread 2V:1H).

Example 1

Unreinforced strip footing: $q_0 = 300$ kPa. Reinforced bed $u/B = 0.4$, $h/B = 0.4$, $N = 4$, $b/B = 5 \rightarrow$ BCR (chart) ≈ 2.2

$q_R = 2.2 \times 300 = 660$ kPa; $q_{all} = 660 / 3 = 220$ kPa (instead of 100 kPa) → footing width can be reduced by ≈ 55 %.

3.3 Pseudo-cohesion approach (idea)

Treat the reinforced zone as soil with apparent cohesion $c_R = (T / 2S_v)\sqrt{K_p}$ (Unit I) and use a two-layer bearing capacity analysis; **use the tension mobilised at small strain (2–5 %)**, not T_{ult}. Numerical (FEM) and Binquet–Lee design charts are the usual practical tools.

Example 2

Sand $\varphi = 35°$: $K_p = 3.69$, $\sqrt{K_p} = 1.92$; geogrid mobilised $T = 10$ kN/m at $S_v = 0.4$ m →

$c_R = (10 / 0.8) \times 1.92 = 24$ kPa (equivalent apparent cohesion of the reinforced zone).

3.4 Settlement behaviour and design checks

- Load–settlement curves of reinforced beds are **stiffer**; the improvement appears mostly **after** the settlement exceeds $\approx$ 2–3 % of B, when the layers are mobilised.

- For design, limit settlement to the permissible value (e.g. 25–50 mm isolated footing) and find q from the reinforced curve at that settlement; apply FS only to the ultimate value.
- Reinforced layers should be **extended and anchored** beyond the footing edges; avoid tearing at the footing edge by using a thin sand cushion for geogrids on angular aggregate.
- In **cohesive soils** replace 1–2 m of weak soil with a compacted granular reinforced bed (sand/gravel mattress with 3–4 geogrid layers); provide drainage.

Example 3 (tension)

Tension check: footing load $q = 220$ kPa on $B = 2$ m; layer at depth $z = 1.0$ m; stress spread 2V:1H → $\Delta\sigma_v = qB/(B + z) = 220 \times 2/3 = 147$ kPa

Assume lateral thrust ≈ $K_a \Delta\sigma_v S_v = 0.27 \times 147 \times 0.8 = 31.8$ kN/m → choose geogrid with $T_{al} \geq 32$ kN/m (i.e. $T_{ult} \approx 32 \times 2.4 \approx 77$ kN/m for PET).

Applications of reinforced soil beds

- Shallow footings, rafts and **tank / silo foundations** on weak sand, silt or fill; embankment and abutment foundations.
- Footings on **expansive / collapsible** soil (with replacement and reinforcement); **machine foundations**; temporary working platforms.
- Strengthening of **strip footings near slopes** (reinforcement increases edge-distance capacity).

4. Reinforced Pavements

Flexible pavements on **weak subgrade** (CBR < 3–4 %) need thick granular layers. Placing a **geogrid or geotextile** at the subgrade–base interface (or within the base / asphalt) gives a more economical and longer-lasting pavement.

4.1 Benefits of reinforcement in flexible pavement layers

1. **Lateral restraint of base aggregate** – interlock with geogrid reduces spreading of aggregate, increasing base stiffness and modulus.

2. **Improved bearing capacity** – failure surface moves into the base; stress on subgrade is reduced (N_c rises from π to $\pi + 2$).
3. **Tensioned-membrane effect** – with rutting, the curved sheet carries load.
4. **Separation / filtration** – prevents subgrade fines pumping into aggregate and keeps the design thickness (geotextile).
5. **Reduced thickness** – base reduction **30–50 %** (aggregate thickness reduction, ATR) for same traffic.
6. **Longer life – Traffic Benefit Ratio TBR** = $N_{reinforced} / N_{unreinforced}$ (typically 1.5 – 10).
7. **Reduced rutting** and **fatigue / reflective cracking** (reinforcement in asphalt layers, geogrid / glass-fibre grid).
8. **Construction platform** on very soft soils (haul roads, working platforms); faster construction; lower life-cycle cost and carbon.

4.2 Types of geosynthetics used in pavements

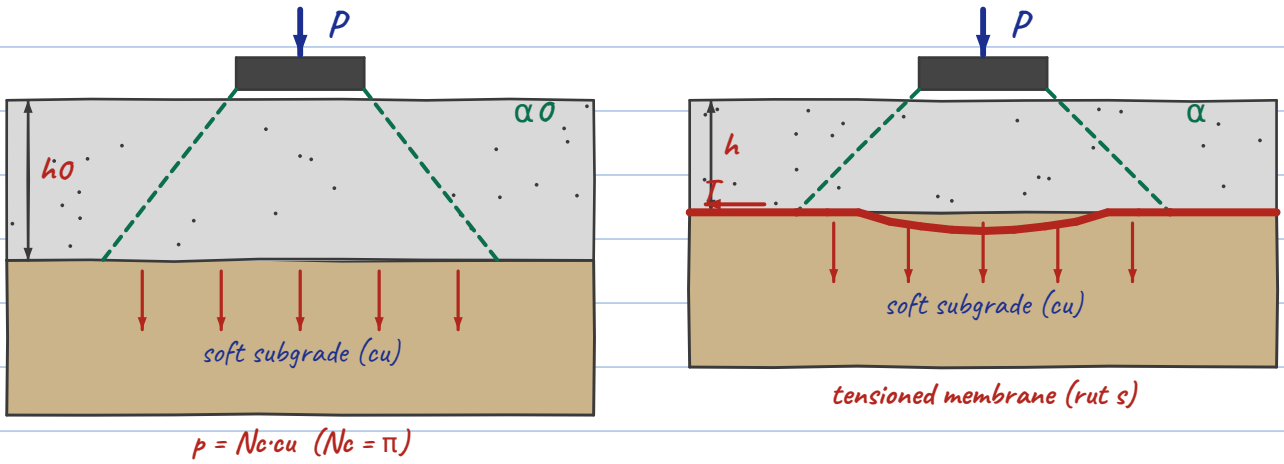
Location	Product	Main function
Subgrade – base interface	Woven GT, biaxial/triaxial geogrid	Separation, reinforcement
Within granular base	Geogrid, geocell	Lateral restraint, stiffness
Under / within asphalt	Glass-fibre, PET or steel grid; paving fabric	Anti-reflective cracking, rutting
Edge / side drain	Nonwoven GT, geocomposite	Filtration, drainage

- Guidelines: **IRC:SP:59** (geotextiles in road pavements), **IRC:37** (flexible pavement design), AASHTO R50 / FHWA manuals.
- Reinforcement is most effective when subgrade CBR is **1–5 %** and traffic is moderate-to-high; for CBR > 8 % benefits are small.
- Benefits are verified by **plate load tests, field trial sections and accelerated pavement testing** (rut depth vs passes).

Example 4

Example benefit calculation: unreinforced base 450 mm; reinforced base 300 mm → ATR = 33 %.

Unreinforced life 10 000 passes; reinforced life 40 000 passes at the same thickness → TBR = 4.



(a) Unreinforced – thicker base (b) Reinforced – thinner base

Fig 5.3 Load spread – unreinforced and reinforced base

Positions: (a) **subgrade / base interface** – most common; (b) **within the base** (upper third) for stiffening; (c) **within or beneath asphalt** (anti-cracking grid).

For subgrade CBR < 1–2 %, use a high-strength geotextile + geogrid combination.

5. Giroud and Noiray (1981) Approach

Design method for **geotextile-reinforced unpaved / haul roads** (also used for base courses of pavements). Based on **load spread through the aggregate** and the bearing capacity of the subgrade, with **tensioned-membrane** action for the reinforced case.

Assumptions

- Wheel load P on circular contact of radius r ($r = \sqrt{[P/(\pi\, p_c)]}$, p_c = tyre pressure) spreads through the aggregate at **angle** α_0 (unreinforced) and α (reinforced, $\geq \alpha_0$).

- Stress on subgrade $p = P / [\pi (r + h \tan\alpha)^2] = p_c / (1 + h \tan\alpha / r)^2$.
- Allowable subgrade stress: **unreinforced** $p = N_c c_u$, $N_c = \pi$ (3.14); **reinforced** $N_c = \pi + 2$ (5.14) because the geotextile prevents the lateral bulge, plus membrane term Δp.
- Undrained strength of subgrade $c_u \approx 30$ CBR (kPa).

Aggregate thickness

$$h = r [\sqrt{(p_c / (N_c c_u))} - 1] / \tan\alpha$$

- **Unreinforced:** h_0 with $N_c = 3.14$ and α_0 (depends on traffic passes N and allowed rut depth; $\alpha_0 \approx 38°$ for a few passes, smaller for heavy traffic – take from design chart).
- **Reinforced:** h with $N_c = 5.14$ (and α); if the rut is large, the geotextile curvature gives a pressure reduction $\Delta p = T / R_c$ ($R_c \approx b^2 / 8s$ radius of the deformed sheet, rut depth s, width b) and strain $\varepsilon \approx (8/3)(s/b)^2$; hence **required tension** $T = \Delta p \cdot R_c$.
- **Savings** $= h_0 - h$; compare with geotextile cost.

Example 5

Single tyre load $P = 40$ kN, $p_c = 480$ kPa → $r = \sqrt{[40/(\pi \times 480)]} = 0.163$ m.

Subgrade CBR = 1 → $c_u \approx 30$ kPa. Take $\alpha_0 = \alpha = 38°$ (tan = 0.781).

Unreinforced: $N_c c_u = 3.14 \times 30 = 94.2$ kPa; $\sqrt{(480/94.2)} = 2.257$ → $h_0 = 0.163 \times (2.257 - 1)/0.781 =$ **0.26 m**

Reinforced: $N_c c_u = 5.14 \times 30 = 154$ kPa; $\sqrt{(480/154)} = 1.764$ → $h = 0.163 \times 0.764/0.781 =$ **0.16 m**

Saving ≈ 0.10 m (≈ 39 %). (Illustration only – real design uses α from the chart for the required number of passes.)

Design steps (Giroud–Noiray)

1. Find wheel load P, tyre pressure p_c, number of passes N, allowed rut depth s.
2. Find subgrade c_u (CBR, vane, CPT).
3. Compute r; choose α_0, α from charts.
4. Compute h_0 ($N_c = \pi$) and h ($N_c = \pi + 2$).

5. Compute required geotextile **tension T** from rut geometry and strain; select product with $T_{al} \geq T$ and compatible modulus.
6. Check filtration, survivability, overlap / lap (0.3–1 m) and construction details.

Giroud–Han (2004) – modern improvement

Giroud and Han extended the approach to **geogrid-reinforced unpaved roads** using a stiffness-dependent bearing factor N_c and a load-spreading angle that depends on **aperture stability modulus**], base thickness and rut depth; the design is based on required base thickness for given traffic N passes and rut depth.

6. Modified CBR Method

The **CBR method** (U.S. Corps of Engineers) gives the pavement / base thickness from the wheel load, tyre pressure and subgrade CBR. In the **modified CBR method** for reinforced pavements, the geosynthetic is accounted for by using a **modified (improved) CBR** of the subgrade, $CBR_m = IF \times CBR_{sg}$, where IF = improvement factor from laboratory CBR tests on reinforced specimens (geosynthetic placed in the mould) or from plate-load / DCP tests; thickness is then read from the standard design curves (IRC:37 / IRC:SP:59) or the formula.

$$t = \sqrt{\{P\,[\,1/(8.1\,CBR) - 1/(\pi\,p)\,]\}}$$ (P in lb, p in psi, t in inch)

$$\text{SI: } t = \sqrt{\{P\,[\,1/(55.9\,CBR) - 1/(\pi\,p)\,]\}}$$ (P in kN, p in kPa, t in m)

Procedure

1. Determine soaked **CBR of subgrade** (unreinforced) – CBR_{sg}.
2. Determine CBR_m with the selected geosynthetic (laboratory reinforced CBR or field test): $IF = CBR_m / CBR_{sg}$ (typically ≈ 1.3–2 for geogrids on weak soil – obtain from tests).
3. Design traffic: wheel load P, tyre pressure p (or MSA for IRC:37).
4. Compute thickness t_0 with CBR_{sg} and t_R with CBR_m.
5. Thickness reduction = $t_0 - t_R$ (percentage savings); check minimum layer thicknesses and drainage.

Example 6

$P = 40$ kN, $p = 480$ kPa, $\pi p = 1508$; $CBR_{sg} = 2$ %, $CBR_m = 3.5$ %

Unreinforced: $1/(55.9 \times 2) = 0.008945$; – $1/1508 = 0.000663 \rightarrow 0.008282 \times 40 = 0.331 \rightarrow t_0 = 0.576$ m

Reinforced: $1/(55.9 \times 3.5) = 0.005111 - 0.000663 = 0.004448 \times 40 = 0.178 \rightarrow t_R = 0.422$ m

Saving = 0.154 m ≈ 27 % of the thickness; IF = 1.75.

Giroud–Noiray vs Modified CBR

Point	Giroud–Noiray	Modified CBR
Basis	Load spread + bearing capacity	Empirical CBR equation / charts
Subgrade property	cu (≈ 30 CBR)	CBR (modified CBRm)
Reinforcement effect	Nc π → π+2, membrane	Improvement factor IF
Product data	Tension, modulus	Test-derived IF
Use	Unpaved / haul roads	Paved flexible pavements

Construction notes

- Trim and compact the subgrade; remove sharp objects; roll out the geogrid / geotextile **flat and taut**; lap **0.3–1 m** (CBR-dependent) or join mechanically.
- Spread aggregate by **end-dumping from the edge** onto the sheet; first lift ≥ 150–300 mm; no turning of tracked vehicles on the bare sheet; compact to ≥ 98 % MDD / specification.
- Provide edge anchorage and drainage; in asphalt reinforcement, use tack coat and compatible grid temperature rating.

Formula sheet – Unit V

Item	Expression
Bearing capacity ratio	$BCR = qR / q0$
Layer depth	$d = u + (N - 1)\,h$
Subgrade stress	$p = P / \pi\,(r + h \tan\alpha)^2$
Aggregate thickness	$h = r(\sqrt{(pc/Nc\ cu)} - 1)/\tan\alpha$
Nc	π (unreinforced); $\pi + 2$ (reinforced)
CBR thickness (SI)	$t = \sqrt{\{P[1/(55.9\ CBR) - 1/(\pi\ p)]\}}$

Important questions

1. What is a reinforced soil bed? Explain mechanisms of improvement of bearing capacity.
2. Discuss the factors affecting the behaviour of reinforced soil beds (u/B, N, h/B, b/B).
3. List the benefits of placing reinforcement in flexible pavement layers.
4. Explain Giroud and Noiray design approach for reinforced pavements with an example.
5. Explain the modified CBR method of design of reinforced pavements.